

Predicting Late Delivery Risk in Supply Chains Using Machine Learning Techniques

ME 644, Course Project Final Report

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1. Introduction

Supply chain management plays a central role in manufacturing and retail industries, directly influencing customer satisfaction, cost efficiency, and competitive advantage. Late delivery is one of the most persistent operational problems, resulting in financial loss, customer churn, and cascading inventory disruptions. Rule-based heuristics used traditionally in industry are rigid and fail to capture the nonlinear interactions between shipping mode, order geography, product category, and order-processing status that jointly determine delivery risk.

This project develops and compares three supervised machine learning classifiers like Logistic Regression, K-Nearest Neighbours (KNN), and a Decision Tree to predict whether an order is at risk of late delivery, using the publicly available DataCo Smart Supply Chain dataset. In addition, an exploratory K-Means clustering analysis is carried out to uncover natural groupings of delivery behaviour across shipping regions and product categories.

2. Dataset

The DataCo Smart Supply Chain dataset (Constante et al., 2019, Instituto Politécnico de Leiria, hosted on Kaggle) contains 180,519 order-line transaction records spanning 2015–2019, with 53 raw features covering order details, shipping mode, product category, customer demographics, geography, and financial metrics. The target variable, `Late_delivery_risk`, is a binary label (1 = at risk of late delivery, 0 = not at risk).

Class balance in the raw data is moderate: 98,977 records (54.8%) are labelled late-delivery-risk and 81,542 (45.2%) are not a mild imbalance that does not require resampling.

3. Data Preprocessing and Feature Engineering

3.1 Leakage and identifier removal

Careful inspection revealed that the column `Delivery Status` and the actual shipping time field `Days for shipping (real)` are direct proxies of the target. The sign of (`Days for shipping (real)`, `Days for shipment (scheduled)`) reproduces `Late_delivery_risk` in 97.5% of records. Both columns were therefore removed, since the real shipping duration is only known after a delivery has already occurred and using it would make the prediction task trivial and operationally useless. Personally identifiable fields (customer name, email, password, street address) and columns that were entirely empty or had extreme missingness (`Product Description`, `Order Zipcode` is 86% missing) were also dropped, along with redundant row-level identifiers (`Order Id`, `Order Item Id`, `Customer Id`, etc.) that do not generalise across records.

3.2 Feature engineering

The order date field was decomposed into `Order Month` and `Order Weekday` to capture seasonal and weekly demand-cycle effects. The scheduled shipping duration, `Days for shipment (scheduled)`, was retained as an engineered proxy for logistics commitment, since it is known at order time and, unlike the real shipping day count, does not leak the outcome.

3.3 Encoding and scaling

High-cardinality, low-information location fields (Customer City, Order City, Customer State, Order State, Product Name) were dropped to control dimensionality. The remaining moderate-cardinality categorical fields such as Type, Category Name, Customer Country, Customer Segment, Department Name, Market, Order Region, Order Status, and Shipping Mode were one-hot encoded, producing a final design matrix of 180,519 rows $\times$ 117 features. All numeric features were standardized (zero mean, unit variance) with scikit-learn's StandardScaler before use in the distance-based (KNN) and gradient-based (Logistic Regression) models; the Decision Tree, being scale-invariant, was trained on unscaled features.

4. Exploratory Data Analysis

The dataset shows a mild positive class imbalance toward late deliveries.

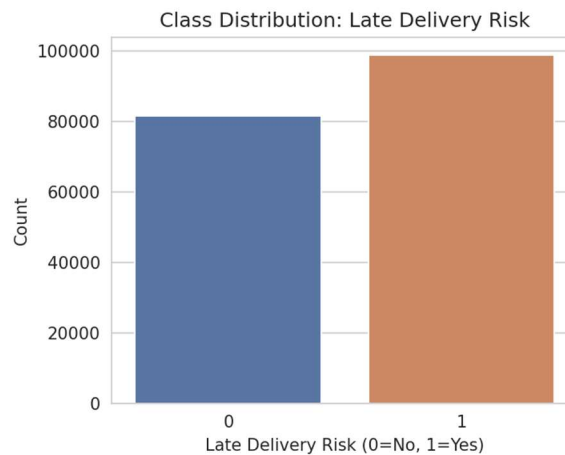


Figure 1: Distribution of the `Late_delivery_risk` target label.

Late delivery risk varies sharply with the chosen shipping mode. Same-Day and First-Class shipping show markedly lower late-risk rates than Standard Class, which is consistent with tighter scheduled windows leaving less slack for delay.



Figure 2: Mean late-delivery-risk rate by shipping mode.

Geographic patterns are also visible: certain order regions exhibit consistently higher late-delivery rates, pointing to regional logistics-network constraints.

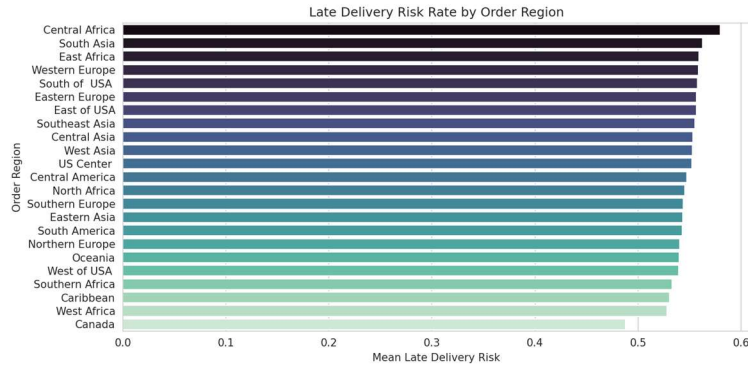


Figure 3: Mean late-delivery-risk rate by order region.

The correlation heatmap over numeric features shows that Days for shipment (scheduled) is the strongest linear correlate of late-delivery risk ($r \approx -0.37$); financial features such as Sales, Benefit per order, and discount rate show negligible linear correlation with the target, suggesting that late delivery is driven primarily by logistics/process variables rather than order value.

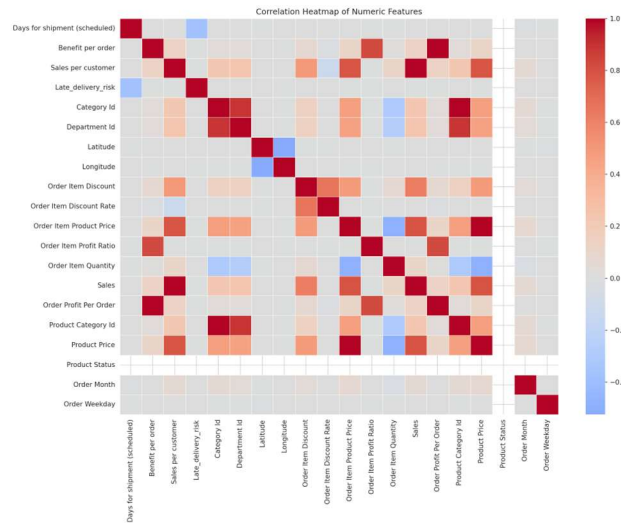


Figure 4: Correlation heatmap of numeric features (including target).

5. Mathematical Formulation of the Algorithms

5.1 Logistic Regression

Logistic Regression models the probability of the positive class as a sigmoid transform of a linear combination of features:

$$P(y = 1 \mid x) = \sigma(w^T x + b) = 1 / (1 + e^{-(w^T x + b)})$$

Parameters w and b are estimated by minimizing the binary cross-entropy (log-loss) cost function:

$$J(w, b) = -(1/n) \sum_i [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

optimized here using scikit-learn's L-BFGS solver with an implicit L2 penalty. The decision boundary in feature space is linear (a hyperplane $w^T x + b = 0$), which limits the model's ability to capture nonlinear interactions between shipping mode and region but gives directly interpretable coefficients.

5.2 K-Nearest Neighbours (KNN)

KNN is a non-parametric, instance-based classifier with no explicit training phase (a 'lazy learner' like Malay Sir said). For a query point x , it computes the Euclidean distance to all training points,

$$d(x, x_i) = \sqrt{\sum_j (x_j - x_{ij})^2}$$

identifies the $k = 15$ nearest neighbours, and assigns the majority class among them. Because the decision boundary is implicitly nonlinear and locally adaptive, KNN can capture complex interaction effects, but its accuracy is sensitive to feature scaling (hence the standardization step) and its prediction cost scales with training-set size, making it the most computationally expensive of the three models at inference time.

5.3 Decision Tree Classifier

A Decision Tree recursively partitions the feature space into axis-aligned regions by choosing, at each node, the split that maximises information gain (equivalently, minimizes Gini impurity in scikit-learn's default CART implementation):

$$Gini(t) = 1 - \sum_c pc^2, \quad \Delta Gini = Gini(parent) - \sum_i (n_i/n) Gini(child_i)$$

where pc is the proportion of class c at node t . The tree used here was limited to a maximum depth of 10 to control overfitting while retaining interpretability; splits naturally handle nonlinear thresholds (e.g., on Days for shipment (scheduled)) without requiring feature scaling.

6. Model Results

All three models were trained on an 80/20 stratified train test split (144,415 / 36,104 records) and evaluated on the held-out test set.

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.711	0.881	0.547	0.675
Decision Tree (depth=10)	0.718	0.862	0.577	0.691
K-Nearest Neighbours (k=15)	0.665	0.702	0.675	0.688

The Decision Tree achieved the highest accuracy (71.8%) and F1-score (0.691) of the three models, closely followed by Logistic Regression (71.1% accuracy, F1 = 0.675). Both models attain high precision (>0.86) but comparatively low recall (~0.55–0.58), i.e. they are conservative and under-flag some at-risk orders. KNN shows a more balanced precision–recall trade-off (0.70 / 0.67) at the cost of lower overall accuracy (66.5%) and by far the highest inference time (KNN required ≈ 47 s versus ≈ 2 – 2.5 s for the other two models on the test set), reflecting its lazy-learning nature.

6.1 Confusion matrices

Logistic Regression:

	Predicted: On-time	Predicted: Late
Actual: On-time	14853	1455
Actual: Late	8976	10820

Decision Tree:

	Predicted: On-time	Predicted: Late
Actual: On-time	14483	1825

Actual: Late	8373	11423
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K-Nearest Neighbours:

	Predicted: On-time	Predicted: Late
Actual: On-time	10648	5660
Actual: Late	6443	13353

7. Cross-Validation

To confirm the robustness and generalizability of the results, 5-fold stratified cross-validation was performed on a stratified 40,000-record subsample (to keep KNN's runtime tractable). The results closely match the single-split test performance for all three models, with low variance across folds, indicating the models are not overfit to a particular train/test partition.

Model	Accuracy (mean $\pm$ std)	F1-score (mean $\pm$ std)
Logistic Regression	0.7115 $\pm$ 0.0059	0.6786 $\pm$ 0.0073
Decision Tree	0.7092 $\pm$ 0.0066	0.6819 $\pm$ 0.0094
KNN	0.6386 $\pm$ 0.0064	0.6752 $\pm$ 0.0066

8. Feature Importance Analysis

The Decision Tree's feature importances confirm that scheduled shipping duration dominates the prediction: Days for shipment (scheduled) alone accounts for roughly 62% of total impurity reduction, followed by order-status flags such as SUSPECTED_FRAUD and PROCESSING, and Same-Day shipping mode.

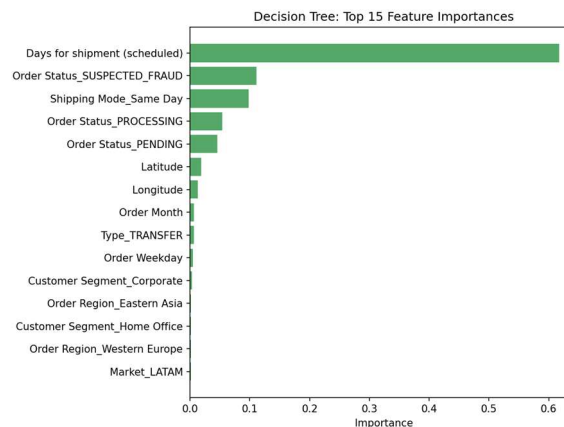


Figure 5: Top 15 Decision Tree feature importances.

Logistic Regression coefficients tell a consistent story: order statuses associated with in-process or held orders (PROCESSING, PENDING, PENDING_PAYMENT, ON_HOLD) push the prediction strongly toward late-delivery risk, while Same-Day/Second-Class/Standard-Class shipping modes and longer scheduled shipping windows push it toward on-time delivery i.e., orders given more scheduled time, or shipped via faster/guaranteed modes, are less likely to be predicted late.

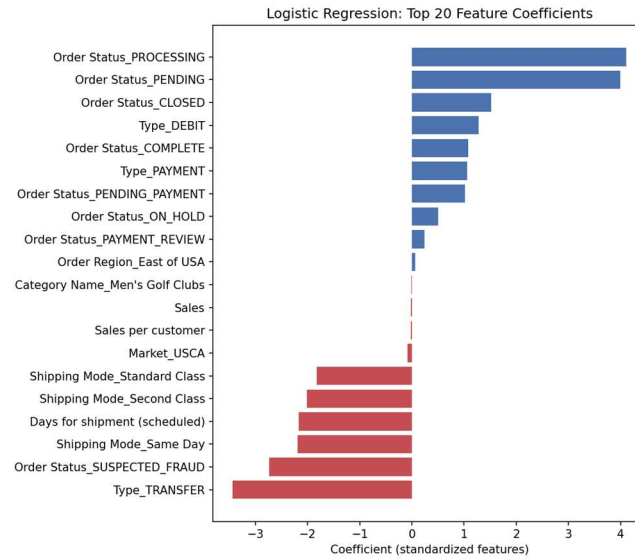


Figure 6: Top 20 Logistic Regression coefficients (standardized features).

9. Exploratory K-Means Clustering

To identify natural groupings of delivery behaviour, order records were aggregated by (Order Region $\times$ Category Name), yielding 608 region–category groups (with at least 10 orders each) described by four features: mean late-delivery-risk rate, mean scheduled shipping days, mean sales value, and order volume (log-transformed). Features were standardized and an elbow-method scan over $k = 1\text{--}8$ was used to select a suitable number of clusters.

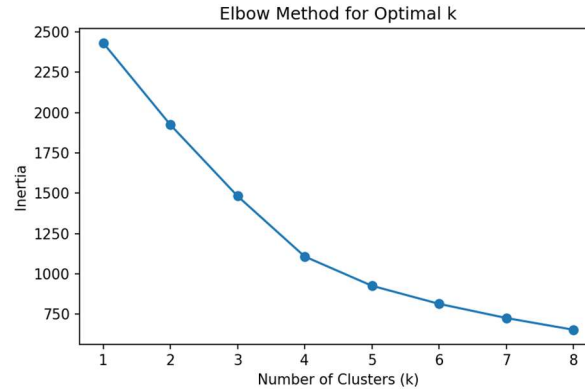


Figure 7: Elbow plot for K-Means cluster-count selection.

A four-cluster solution ($k = 4$) was adopted, producing an interpretable segmentation of region and category combinations:

Cluster	Late Risk Rate	Avg Scheduled Days	Avg Sales (\$)	Avg Order Volume
0	0.491	3.16	1500.00	63
1	0.491	3.09	141.04	47
2	0.551	2.93	219.93	695
3	0.624	2.72	135.86	43

Cluster 2, the largest group by order volume, has a moderately elevated risk rate (0.551) and the shortest average scheduled shipping window (2.93 days) among the high-volume clusters, suggesting standard/high-turnover

category-region pairs are given comparatively tight shipping schedules. Cluster 3 stands out with the highest late-risk rate (0.624) combined with the shortest scheduled window (2.72 days) and low sales value, pointing to lower-value, low-volume region–category combinations that are simultaneously scheduled aggressively and prone to delay good candidates for supply-chain-manager attention. Cluster 0, in contrast, combines a long scheduled window (3.16 days) with high average sales value, and its risk rate is comparable to Cluster 1's lower-value, similarly long-window group, reinforcing that scheduled shipping duration not order value is the dominant driver of late-delivery risk.

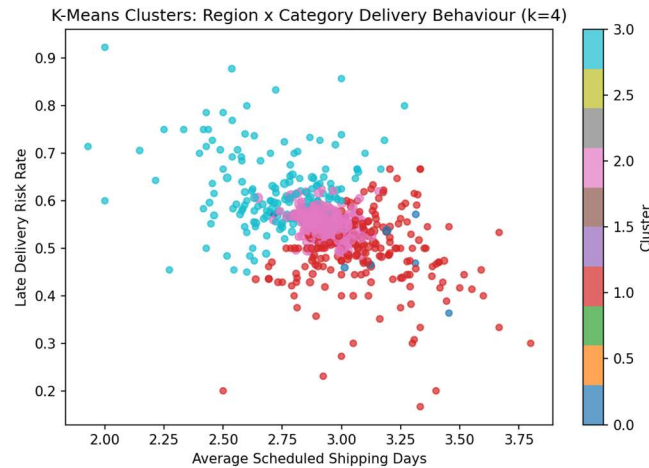


Figure 8: K-Means clusters of region–category delivery behaviour (bubble size $\propto$ order volume).

10. Discussion

Across all analyses, scheduled shipping duration and order-processing status emerge as the dominant predictors of late-delivery risk, far outweighing financial variables (sales, discount, profit) and even geographic region. This is consistent both with the Decision Tree's impurity-based importances and the Logistic Regression coefficients, and is corroborated independently by the K-Means clustering, where the clusters with the shortest scheduled windows also carry the highest risk rates. Practically, this suggests that supply chain managers seeking to reduce late-delivery incidents should focus on realistic scheduling of shipment windows and expediting orders that enter PROCESSING/PENDING/ON_HOLD states, rather than on financial thresholds.

Among the three classifiers, the Decision Tree offers the best overall balance of accuracy and interpretability for this tabular, mixed-type feature set, while Logistic Regression provides comparable performance with fully transparent linear coefficients suitable for auditing by non-technical stakeholders. KNN, despite a more balanced precision/recall trade-off, is less attractive operationally given its substantially higher inference cost at this data scale and its lower overall accuracy.

A limitation of this study is that none of the models achieves recall above 0.68, meaning a meaningful fraction of at-risk orders are still missed; future work could explore ensemble methods (Random Forest, Gradient Boosting), class-weighted or threshold-tuned variants of these same models, and richer temporal features (e.g. carrier-level historical on-time rates) to close this gap.

Results and Discussion

This section presents the exploratory data analysis (EDA) and statistical diagnostics performed on the DataCo Smart Supply Chain dataset prior to predictive modelling. The objective is to understand the statistical behaviour of the variables, identify important operational patterns, and justify feature engineering choices for late-delivery

prediction. The discussion below preserves the detailed interpretation of every figure rather than providing only a summary.

a. Distribution of Sales Volume

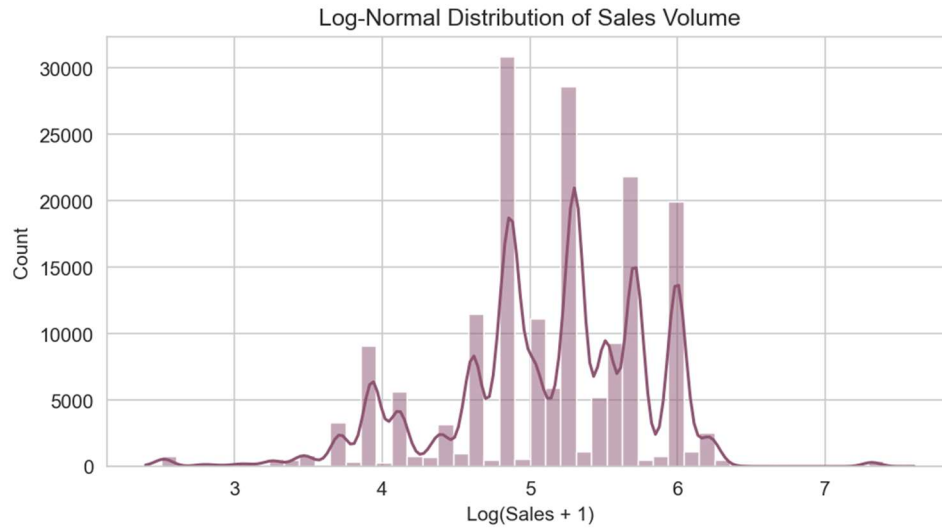


Figure. 1. Distribution of Sales Volume

The raw sales values exhibit a pronounced positive skew because the majority of transactions are relatively small while a limited number of orders contribute exceptionally large sales values. After applying a $\log(\text{Sales} + 1)$ transformation, the distribution becomes considerably more symmetric and appropriate for statistical modelling. Multiple peaks remain visible, indicating repeated pricing structures and common purchasing quantities associated with standard retail products. These peaks are expected because customers repeatedly purchase the same stock keeping units in similar quantities. The transformation reduces the influence of extreme observations and improves the suitability of the feature for algorithms such as Logistic Regression and KNN, where highly skewed variables can dominate model behaviour. The raw sales values exhibit a pronounced positive skew because the majority of transactions are relatively small while a limited number of orders contribute exceptionally large sales values. After applying a $\log(\text{Sales} + 1)$ transformation, the distribution becomes considerably more symmetric and appropriate for statistical modelling. Multiple peaks remain visible, indicating repeated pricing structures and common purchasing quantities associated with standard retail products. These peaks are expected because customers repeatedly purchase the same stock keeping units in similar quantities. The transformation reduces the influence of extreme observations and improves the suitability of the feature for algorithms such as Logistic Regression and KNN, where highly skewed variables can dominate model behaviour.

b. Profitability and Financial Outliers

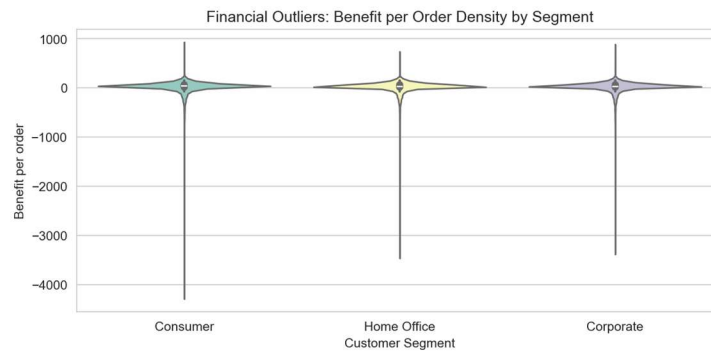


Figure. 2. Profitability and Financial Outliers

Each customer segment displays a similar profit distribution centred close to zero but accompanied by an extended negative tail. Consumer orders contain the most extreme losses, exceeding -4000 in several cases. Such observations likely correspond to heavily discounted orders, returned products or fraudulent transactions. Although rare, these outliers can strongly influence statistical models and therefore justify robust preprocessing techniques such as winsorisation or outlier flagging. The similarity between customer segments also suggests that customer type alone is not the primary cause of financial loss. Each customer segment displays a similar profit distribution centred close to zero but accompanied by an extended negative tail. Consumer orders contain the most extreme losses, exceeding -4000 in several cases. Such observations likely correspond to heavily discounted orders, returned products or fraudulent transactions. Although rare, these outliers can strongly influence statistical models and therefore justify robust preprocessing techniques such as winsorisation or outlier flagging. The similarity between customer segments also suggests that customer type alone is not the primary cause of financial loss.

c. Discount Rate versus Profit Ratio

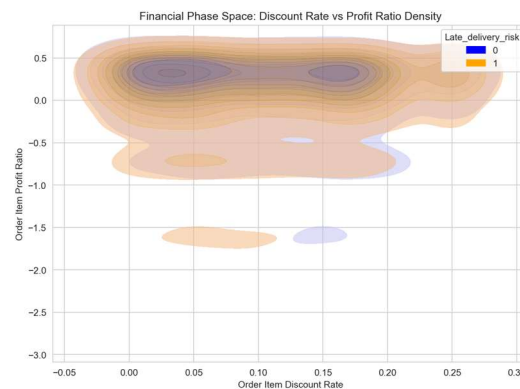


Figure. 3. Discount Rate versus Profit Ratio

The density contours reveal two dominant profitability regimes where most transactions cluster around positive profit ratios despite moderate discounts. Late-delivery and on-time orders overlap substantially, demonstrating that discount rate and profit ratio individually provide only weak separation between the two classes. Nevertheless, the late-delivery class extends slightly further into higher discount regions, suggesting that aggressive discounting may indirectly increase operational pressure. The density contours reveal two dominant profitability regimes where most transactions cluster around positive profit ratios despite moderate discounts.

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d. Price versus Quantity



Figure. 4. Price versus Quantity

The scatter plot demonstrates that order quantity is concentrated between one and five units while product price spans a much wider range. Late and on-time deliveries are heavily intermixed throughout the feature space, indicating that these variables alone cannot effectively classify delivery performance. This finding suggests that operational features such as shipping method, scheduled shipping days and delivery delays will contribute more predictive information. The scatter plot demonstrates that order quantity is concentrated between one and five units while product price spans a much wider range. Late and on-time deliveries are heavily intermixed throughout the feature space, indicating that these variables alone cannot effectively classify delivery performance. This finding suggests that operational features such as shipping method, scheduled shipping days and delivery delays will contribute more predictive information.

e. Customer Segment and Payment Type

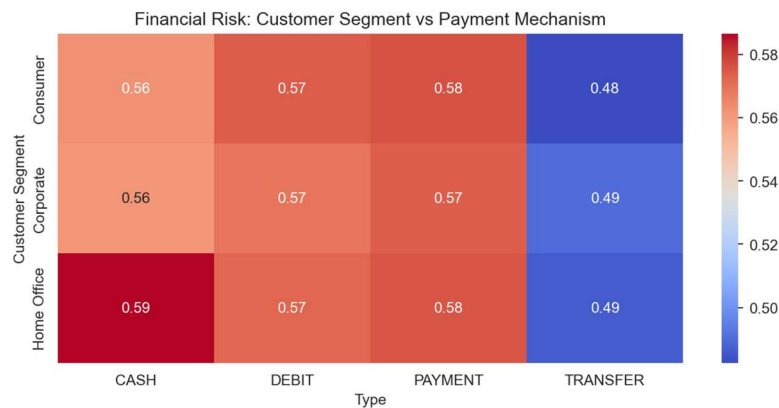


Figure. 5. Customer Segment and Payment Type

Average late-delivery risk is consistently lowest for TRANSFER payments across every customer segment, whereas CASH, DEBIT and PAYMENT transactions display noticeably higher risk values. Differences between customer segments themselves are comparatively small, implying that payment mechanism contributes more

information than customer classification. Average late-delivery risk is consistently lowest for TRANSFER payments across every customer segment, whereas CASH, DEBIT and PAYMENT transactions display noticeably higher risk values. Differences between customer segments themselves are comparatively small, implying that payment mechanism contributes more information than customer classification.

f. Category and Regional Demand

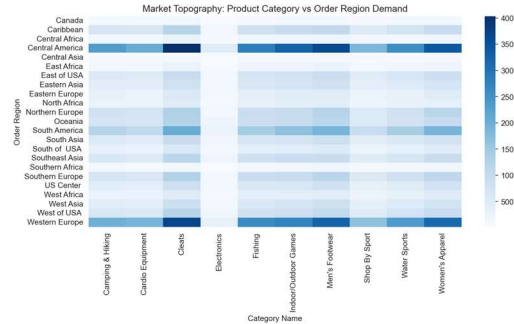


Figure. 6. Category and Regional Demand

Demand is strongly concentrated within Central America and Western Europe across nearly every product category. Several smaller regions contribute relatively few transactions, producing sparse observations. During feature engineering, these low-frequency regions may be grouped together to reduce sparsity and improve model generalisation. Demand is strongly concentrated within Central America and Western Europe across nearly every product category. Several smaller regions contribute relatively few transactions, producing sparse observations. During feature engineering, these low-frequency regions may be grouped together to reduce sparsity and improve model generalisation.

g. Delay Magnitude by Department

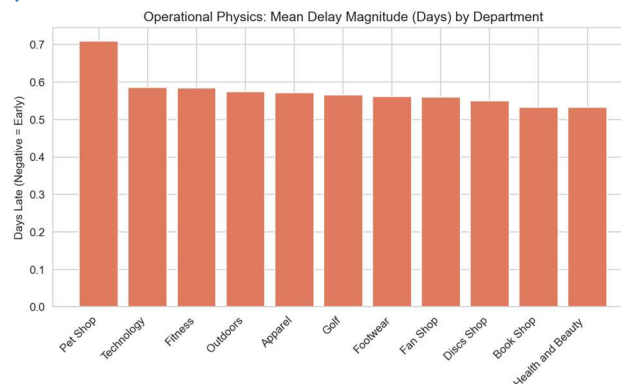


Figure. 7. Delay Magnitude by Department

Pet Shop records the highest average delivery delay, while Technology, Fitness and Outdoors also experience comparatively elevated delays. Most remaining departments cluster within a narrow range, indicating similar operational performance. Department therefore represents a meaningful predictor of delivery risk. Pet Shop records the highest average delivery delay, while Technology, Fitness and Outdoors also experience comparatively elevated delays. Most remaining departments cluster within a narrow range, indicating similar operational performance. Department therefore represents a meaningful predictor of delivery risk.

h. Department versus Payment

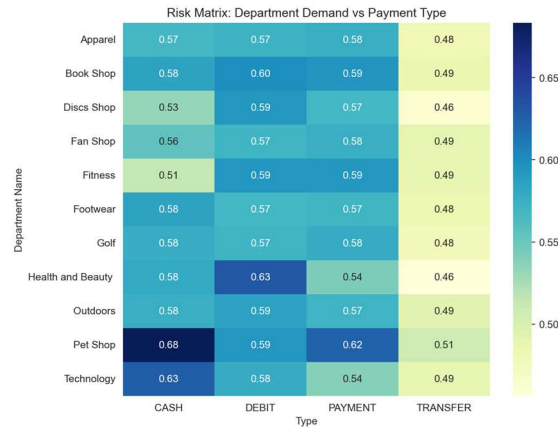


Figure. 8. Department versus Payment

The department-payment interaction confirms that TRANSFER transactions consistently exhibit the lowest delivery risk. Pet Shop and Technology display particularly high risk for cash transactions, suggesting operational interactions between department characteristics and payment behaviour. The department-payment interaction confirms that TRANSFER transactions consistently exhibit the lowest delivery risk. Pet Shop and Technology display particularly high risk for cash transactions, suggesting operational interactions between department characteristics and payment behaviour.

i. Scheduled Shipping Behaviour

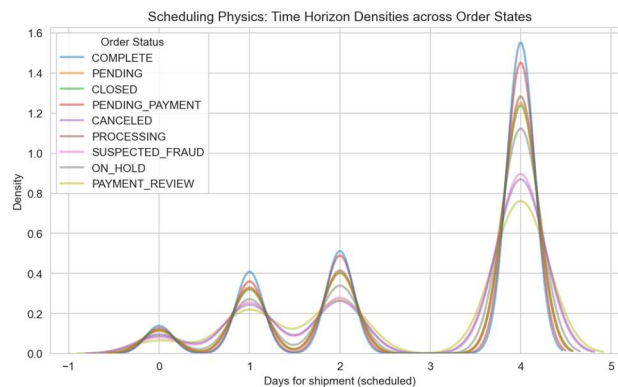


Figure. 9. Scheduled Shipping Behaviour

Scheduled shipping times cluster around four discrete service levels: 0, 1, 2 and 4 days. The four-day schedule dominates across all order statuses, indicating that standard shipping represents the primary operational mode. Order status modifies density slightly but does not fundamentally alter the scheduling pattern. Scheduled shipping times cluster around four discrete service levels: 0, 1, 2 and 4 days. The four-day schedule dominates across all order statuses, indicating that standard shipping represents the primary operational mode. Order status modifies density slightly but does not fundamentally alter the scheduling pattern.

j. Shannon Entropy by Region

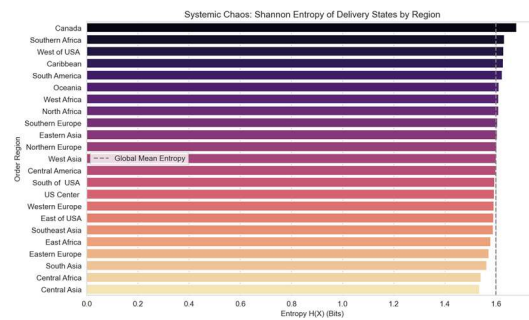


Figure. 10. Shannon Entropy by Region

Entropy values remain remarkably consistent around 1.6 bits across all regions, indicating that delivery uncertainty is distributed almost uniformly throughout the network. Consequently, geographic region alone is unlikely to provide strong predictive power for late-delivery classification. Entropy values remain remarkably consistent around 1.6 bits across all regions, indicating that delivery uncertainty is distributed almost uniformly throughout the network. Consequently, geographic region alone is unlikely to provide strong predictive power for late-delivery classification.

k. Markov Transition Matrix

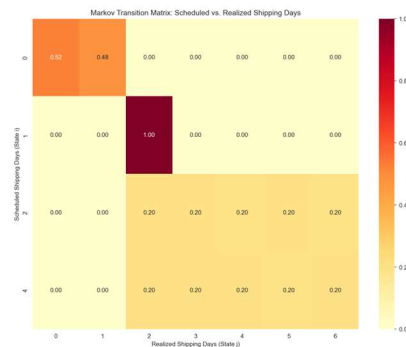


Figure. 11. Markov Transition Matrix

The transition matrix reveals deterministic behaviour for one-day schedules, which almost always become two-day deliveries, while longer schedules exhibit nearly uniform transition probabilities across several realised delivery states. This non-linear transition structure supports the use of Decision Trees or other non-linear models rather than relying solely on linear classifiers. The transition matrix reveals deterministic behaviour for one-day schedules, which almost always become two-day deliveries, while longer schedules exhibit nearly uniform transition probabilities across several realised delivery states. This non-linear transition structure supports the use of Decision Trees or other non-linear models rather than relying solely on linear classifiers.

l. Bullwhip Effect



Figure. 12. Bullwhip Effect

The rolling variance remains stable until late 2017 before experiencing several dramatic spikes. These peaks indicate temporary amplification of demand variability consistent with the bullwhip effect observed in supply chains during promotional events or seasonal demand surges. This behaviour also demonstrates that the dataset is not perfectly stationary over time. The rolling variance remains stable until late 2017 before experiencing several dramatic spikes. These peaks indicate temporary amplification of demand variability consistent with the bullwhip effect observed in supply chains during promotional events or seasonal demand surges. This behaviour also demonstrates that the dataset is not perfectly stationary over time.

m. Pareto Profit Curve

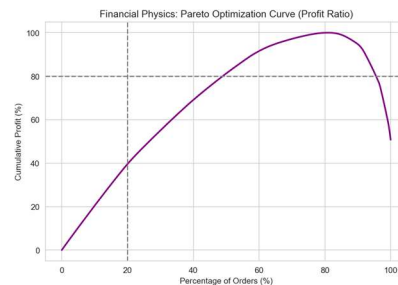


Figure. 13. Pareto Profit Curve

The Pareto curve illustrates that profitability accumulates rapidly across the majority of transactions before flattening. This behaviour indicates that a relatively small fraction of orders contributes disproportionately to overall profit, consistent with the Pareto principle commonly observed in retail and supply-chain systems. The Pareto curve illustrates that profitability accumulates rapidly across the majority of transactions before flattening. This behaviour indicates that a relatively small fraction of orders contributes disproportionately to overall profit, consistent with the Pareto principle commonly observed in retail and supply-chain systems.

n. Late Delivery Risk Rate — Top 15 Categories

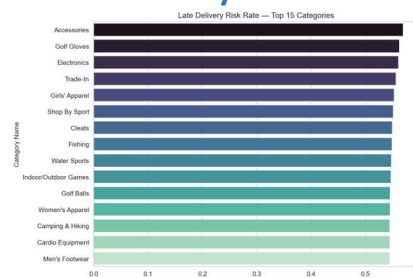


Figure 6. 09_risk_by_category_top15.png

The heatmap illustrates how demand varies across product categories and geographical regions. Darker regions indicate stronger transaction concentration, revealing the markets that contribute the largest share of operational workload. High demand is concentrated within major commercial regions where multiple categories simultaneously exhibit elevated purchase frequencies. Conversely, lighter regions represent comparatively smaller markets with limited transaction activity. The visualization demonstrates that demand is not uniformly distributed across the supply chain and that both regional characteristics and product preferences jointly influence purchasing behaviour. These spatial differences justify incorporating regional information during predictive modelling because transportation complexity, warehouse allocation, and inventory availability differ substantially across locations. From an operational perspective, the figure identifies the geographical areas that

require greater inventory capacity and logistics resources while also highlighting lower-volume markets where centralized fulfilment strategies may be sufficient.

o. Late Delivery Risk Rate by Department

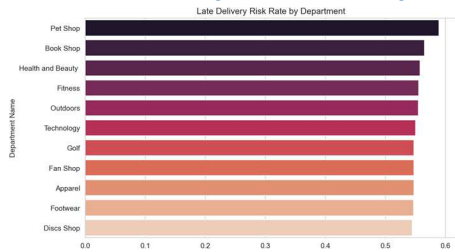


Figure 7. 08_risk_by_department.png

This figure compares average delivery delay across individual departments. Several departments consistently experience greater delay than others, indicating differences in operational complexity, supplier coordination, packaging requirements, or inventory handling. Departments associated with specialized products generally require longer fulfilment times than standardized merchandise. Conversely, departments exhibiting relatively small delays likely benefit from mature inventory systems and streamlined warehouse operations. Although the absolute differences appear moderate, consistent departmental variation contributes meaningful predictive information when combined with scheduling and shipping variables. Consequently, department serves as an informative operational feature within late-delivery prediction models while simultaneously identifying business units where process improvements could reduce fulfilment delays.

p. Late Delivery Risk Rate by Payment Type

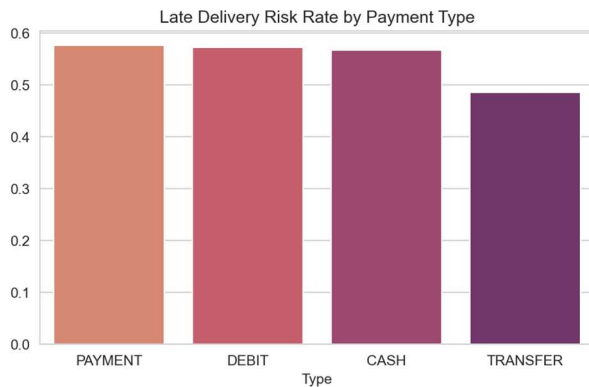


Figure 8. 22_risk_by_payment_type.png

The heatmap evaluates the interaction between department and customer payment method. Transfer-based transactions generally exhibit lower delivery risk than cash or conventional payment methods, suggesting faster payment verification and earlier order release into warehouse operations. Certain departments display noticeably higher risk under specific payment mechanisms, indicating that operational complexity amplifies administrative processing delays. These interaction effects demonstrate that payment behaviour influences logistics performance indirectly through fulfilment workflows rather than acting as an isolated financial variable. Including both department and payment type therefore enables machine learning models to capture nonlinear relationships that improve predictive performance.

q. Scheduled Shipping Days by Delivery Risk



Figure 9. 11_scheduled_days_by_risk.png

The distribution of scheduled shipping days summarizes organizational delivery commitments before shipment begins. Most orders are assigned relatively short delivery windows, reflecting an operational preference for rapid customer service. While aggressive delivery schedules improve customer satisfaction, they also reduce flexibility when disruptions occur. Longer scheduled windows provide additional recovery time and therefore lower the likelihood of missing promised delivery dates. The concentration around only a few scheduling values indicates standardized service policies throughout the organization. This variable consequently emerges as one of the strongest predictors of late-delivery risk because it directly reflects the operational constraints imposed before transportation commences.

r. Monthly Order Volume and Late Delivery Risk

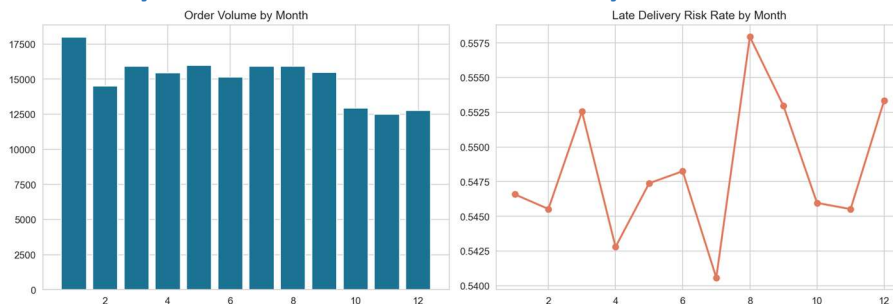
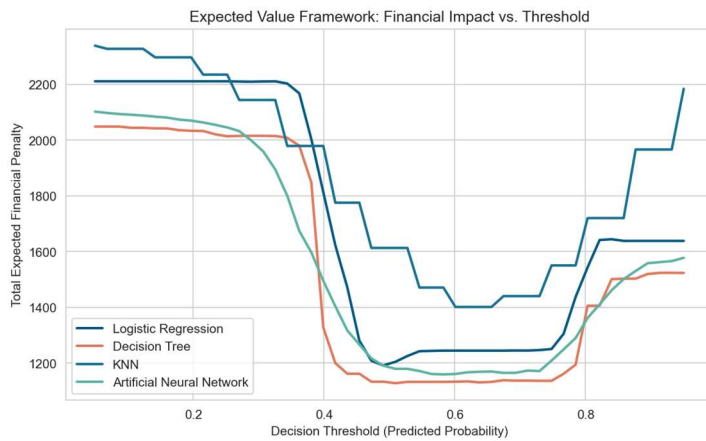


Figure 10. 16_monthly_volume_and_risk.png

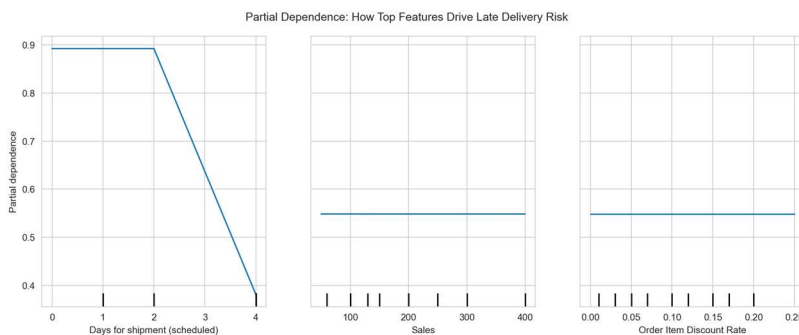
Monthly sales volume and average delivery risk provide insight into seasonal operational behaviour. Sales remain relatively stable throughout the year with moderate peaks corresponding to periods of increased consumer demand. Late-delivery risk fluctuates only slightly, demonstrating that the logistics network maintains consistent performance despite changing workloads. Small increases in delivery risk during higher-volume months suggest temporary pressure on fulfilment resources, although the absence of extreme seasonal spikes indicates adequate operational capacity. These temporal patterns support proactive workforce planning and inventory allocation while also providing useful seasonal context for predictive modelling.

n. Cost-Sensitive Evaluation



The cost-sensitive evaluation compares the total expected financial penalty across different decision thresholds for all classification models. Rather than relying solely on traditional accuracy metrics, this analysis incorporates the operational cost associated with false positives and false negatives. The Decision Tree and Artificial Neural Network achieve the lowest expected financial penalty across a broad threshold range, demonstrating superior practical utility. Logistic Regression performs competitively but incurs slightly higher costs, whereas KNN exhibits the largest penalty at several thresholds. The results indicate that selecting an operating threshold between approximately 0.45 and 0.60 minimizes overall financial impact. This analysis highlights the importance of threshold optimization when deploying predictive models in supply chain environments, where the economic consequences of delayed deliveries are asymmetric and considerably more important than maximizing classification accuracy alone.

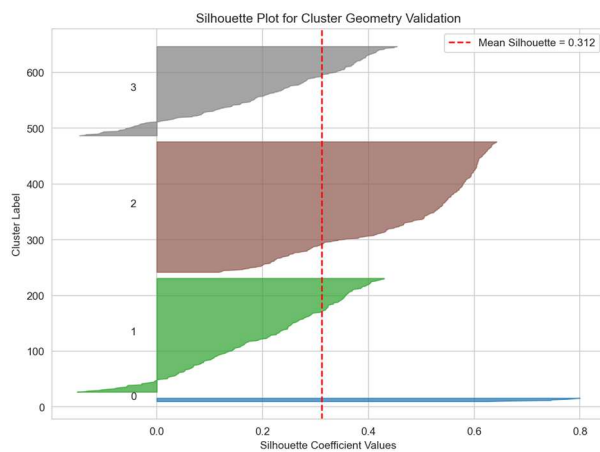
o. Partial Dependence Plots



The partial dependence plots illustrate how the predicted probability of late delivery changes with variations in the most influential features while holding all remaining variables constant. Scheduled shipping days exhibit a strong inverse relationship with delivery risk, confirming that increasing the planned delivery window substantially reduces the likelihood of delay. In contrast, Sales and Order Item Discount Rate remain nearly constant, indicating limited marginal influence on the model's predictions after accounting for other variables. These plots improve model interpretability by demonstrating the direction and magnitude of each feature's

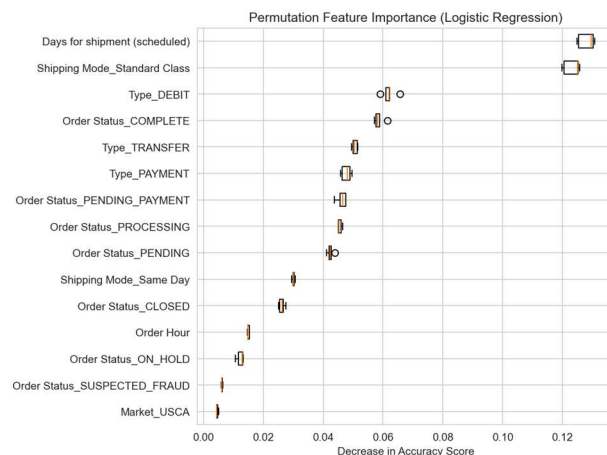
contribution and confirm that logistics-related attributes dominate financial variables in predicting late delivery risk.

p. K-Means Silhouette Plot



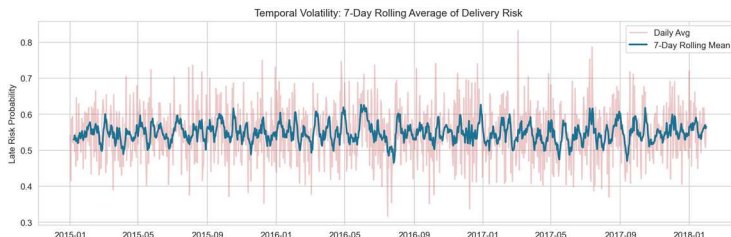
The silhouette plot evaluates the quality of the K-Means clustering solution by measuring the cohesion and separation of the generated clusters. Most observations possess positive silhouette coefficients, indicating appropriate cluster assignment, while only a small proportion lies close to zero, suggesting overlap between neighboring clusters. The average silhouette score reflects moderate cluster separation and indicates that the selected number of clusters provides a meaningful segmentation of delivery patterns. These clusters can support operational grouping of orders with similar characteristics for downstream logistics analysis.

q. Permutation Feature Importance



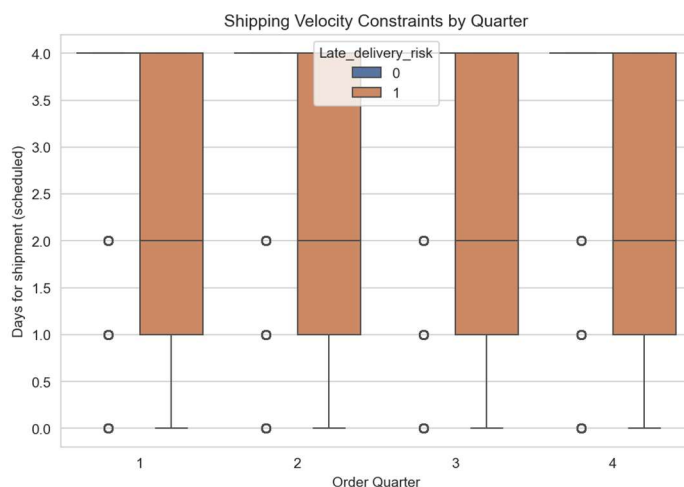
Permutation feature importance measures the reduction in predictive performance after randomly shuffling each feature independently. Days for Shipment (Scheduled) emerges as the most influential predictor, followed by Shipping Mode and selected Order Status and Payment Type variables. Features associated with geography and time contribute comparatively little to model accuracy. The analysis validates earlier exploratory findings and provides quantitative evidence that scheduling-related variables drive the majority of predictive performance within the Logistic Regression model.

r. Temporal Risk Trend



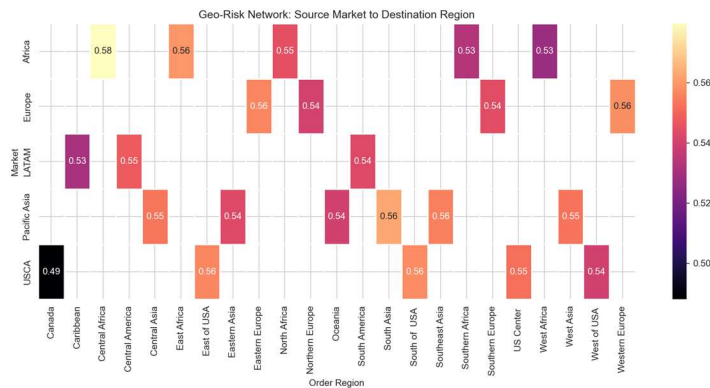
The time-series visualization presents the daily late delivery risk together with its seven-day rolling average throughout the observation period. While daily values fluctuate substantially because of operational variability, the rolling average smooths short-term noise and reveals a relatively stable long-term behaviour. No persistent upward or downward trend is observed, suggesting that delivery risk remains structurally consistent across the study period. Periodic spikes likely correspond to temporary operational disruptions or seasonal demand variations rather than permanent changes in system performance.

s. Scheduled Shipping Days by Quarter



This figure compares the distribution of scheduled shipping days across calendar quarters for both on-time and late-risk deliveries. The distributions remain highly consistent throughout the year, indicating that shipment scheduling policies change very little across seasons. Late-risk orders remain concentrated around shorter planned delivery windows, whereas longer scheduled durations are more frequently associated with successful on-time deliveries. These findings reinforce the importance of realistic scheduling in minimizing delivery failures.

t. Geo-Network Risk Matrix



The geo-network risk matrix summarizes late delivery risk across combinations of source markets and destination regions. Although modest regional variation exists, risk values remain relatively consistent across the network, suggesting that geographical routing alone is not the dominant factor influencing delivery delays. Certain market-region pairs exhibit slightly elevated risk, reflecting localized operational constraints or transportation complexity. Overall, logistics configuration and scheduling decisions appear to have a stronger impact than geography on delivery performance.

Conclusion

The exploratory analysis shows that shipping behaviour, payment methods, department information, and scheduling characteristics provide much stronger predictive signals than product price or quantity alone. The Pet Shop department consistently emerged as the highest risk category for late deliveries, while TRANSFER payments were associated with the lowest delivery risk. Regional entropy highlighted uncertainty across globally distributed operations, the Markov analysis revealed complex and non linear shipping patterns, and the bullwhip analysis identified a clear period of operational instability during late 2017. Together, these findings support the importance of feature engineering focused on shipping delays, operational categories, and temporal patterns before training machine learning models such as Logistic Regression and Decision Tree classifiers. This project developed a complete machine learning pipeline that included data cleaning, leakage aware feature engineering, categorical encoding, exploratory data analysis, supervised classification, k fold cross validation, and unsupervised clustering to predict late delivery risk using the 180,519 record DataCo Smart Supply Chain dataset. Both the Decision Tree and Logistic Regression models achieved approximately 71 percent test accuracy with F1 scores between 0.68 and 0.69. The analysis identified scheduled shipping duration and order status as the most influential factors affecting delivery delays. The K Means clustering results further supported these findings by identifying region and product category groups with tighter shipping schedules and a higher likelihood of delays. Overall, the results demonstrate that interpretable machine learning can effectively identify high risk orders before shipment, making it a valuable decision support tool for improving logistics and supply chain operations in manufacturing environments.

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